

A singular nonproperness component of a five-dimensional Keller map

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Abstract

We study an explicit polynomial map of affine five-space over $\mathbb{Q}(\sqrt{-2})$, obtained by a weighted lift of a mixed polynomial sweep. It has Jacobian determinant one and geometric degree four. An exact squarefree quartic gives four distinct points of one fiber without numerical root approximation. Our main result concerns its nonproperness geometry: we exhibit an irreducible component whose reduced singular locus consists of $\mathbb{G}_m \times \mathbb{A}^2$ and $\mathbb{G}_m \times \mathbb{A}^1$ meeting in $\mathbb{G}_m$. The component itself has Hodge–Deligne polynomial $(uv)^4$, but its singular locus has polynomial $(uv)^3 - 2uv + 1$. We normalize componentwise singular Hodge polynomials by the ambient dimension and prove their invariance under the equivalence relation generated by polynomial left–right changes and genuine identity extensions. The nonzero constant term of the displayed singular polynomial excludes every representative of dimension less than five in this equivalence class. Explicit formulas, full geometric proofs, and a portable exact-arithmetic supplement identify the same polynomial map throughout.

Contribution and significance

The contribution is a specific nonproperness component, its complete reduced singular geometry, and the resulting dimension obstruction for an explicit Keller map. The product behavior of nonproperness loci is background, not a new theorem: it already underlies stable normalization-boundary arguments [11]. Componentwise generic-fiber data have also been used in a neighboring weighted-lift family [8]. Monson’s stable classification uses normalized singular discriminant data in a three-dimensional, degree-three family [9].

Here the decisive decoration is the *intrinsic singular locus*. Its Hodge polynomial detects a stable dimension obstruction invisible to the total Hodge polynomial of the component. The precise additional statement is Theorem 1.1; neither a first counterexample to the Jacobian Conjecture nor a new general product identity is claimed. Complete geometric arguments and exact coefficient data make the dimension obstruction traceable to one fixed map rather than a family-level analogy.

1. Main result and context

All varieties in the geometric statements are reduced varieties over $\mathbb{C}$. The explicit coefficients initially lie in

$$k = \mathbb{Q}(\eta), \quad \eta^2 = -2, \quad \mu = \frac{2+4\eta}{9}, \quad 9\mu^2 - 4\mu + 4 = 0.$$

Either embedding $k \hookrightarrow \mathbb{C}$ is allowed. For a dominant generically finite polynomial map F , write $J(F)$ for its reduced nonproperness set: a target point is outside $J(F)$ if it has a Zariski-open

neighborhood over which F is finite. Polynomial maps F, G in the same dimension are *left–right equivalent* if $G = \beta \circ F \circ \alpha$ for polynomial automorphisms of source and target. Write $F \approx G$ for the equivalence relation generated by these equivalences and the pairs $F \leftrightarrow F \times \text{id}_{\mathbb{A}^1}$. Only genuine identity-extension pairs are generators; deleting an arbitrary coordinate is not an allowed operation.

Theorem 1.1. *The map $\mathcal{F} : \mathbb{A}_k^5 \rightarrow \mathbb{A}_k^5$ defined in Section 2 has the following properties.*

1. *It is polynomial, $\det J(\mathcal{F}) = 1$, and its geometric degree is four. The fiber over $(1, 1, 0, 0, 0)$ contains four distinct geometric points.*
2. *Its complex nonproperness set has an irreducible fourfold component X whose reduced singular locus has two components isomorphic to $\mathbb{G}_m \times \mathbb{A}^2$ and $\mathbb{G}_m \times \mathbb{A}^1$, with intersection $\mathbb{G}_m$.*
3. *Writing $q = uv$, one has*

$$[X] = \mathbb{L}^4, \quad E_c(X; u, v) = q^4, \quad E_c(\text{Sing}(X)_{\text{red}}; u, v) = q^3 - 2q + 1.$$

4. *For any complex variety W and integers $b > a \geq 0$, $X \times \mathbb{A}^a \not\cong W \times \mathbb{A}^b$. Consequently $\mathcal{F} \not\approx G$ for any square polynomial map G of dimension less than five.*

The explicit construction is the degree-one affine-product branch of a mixed sweep associated with Gao’s polynomial Hamilton–Jacobi framework [4]. A companion paper treats the all-degree sweep classification; no result of that companion is needed here. Earlier degree-spectrum constructions [3] address the existence of Keller maps of prescribed geometric degree, while the present question is the dimension of a representative under a specified generated relation. The relation is the complex analogue of the polynomial stable relation defined by Campbell [1, Section 3]; we prove the required common-stabilization lemma below rather than attributing it to that source. It is stronger than merely excluding a product with the number of identity factors needed to reach dimension five, as in a direct stabilization comparison [7].

Jelonek’s work supplies the foundational nonproperness-set setting [5, 6]. Our use of a singular Hodge polynomial does not presume affine cancellation and does not cancel $\mathbb{L}$ in the Grothendieck ring. This distinction matters in stable-isomorphism questions [10]. The theorem excludes lower-dimensional representatives; it does not classify representatives in dimension five.

2. The map and its exact polynomial construction

Set $\kappa = -3\mu/2$, and define

$$\begin{aligned} a(x) &= a_0 + a_1x + a_2x^2, & s(x) &= s_0 + s_1x + s_2x^2, \\ a_0 &= (3\mu - 2)/2, & a_1 &= 2(\mu + 2)/3, & a_2 &= -4(5\mu - 2)/3, \\ s_0 &= (63\mu - 46)/256, & s_1 &= -(45\mu - 2)/64, & s_2 &= (9\mu + 2)/16. \end{aligned}$$

For $\tau = \theta + s(x)$ and $z = y + a(x)$ put

$$\begin{aligned} f &= \tau z - s'(x)/2, \\ g &= \mu\tau z^3 - (a'(x)\tau + s''(x)/2)z + a''(x)\tau/2 + s''(x)a(x)/2 + \kappa. \end{aligned} \tag{2.1}$$

Let $\mathcal{L}_f = \partial_x + 2f\partial_\theta$, and define

$$\begin{aligned} X_1 &= -\frac{1}{2}(g_y + \mathcal{L}_f f), & X_2 &= f + xX_1, \\ X_3 &= 2xf - \theta + x^2X_1, & X_4 &= g + yX_1, \\ S(\gamma, x, y, \theta) &= (X_1, X_2, X_3, X_4) + \gamma(1, x, x^2, y). \end{aligned} \tag{2.2}$$

All the expressions so far are polynomial. In ordered source variables $(\rho, Y_1, Y_2, Y_3, Y_4)$ set

$$v_1 = \rho Y_1, \quad v_2 = \rho^2 Y_2, \quad v_3 = \rho^3 Y_3, \quad v_4 = \rho^3 Y_4. \quad (2.3)$$

The constants for a triangular coordinate change are

$$\begin{aligned} \ell_{21} &= -(9\mu + 26)/16, & \ell_{31} &= -2(11\mu - 14)/3, \\ \ell_{32} &= 2(19\mu - 16)/3, & \ell_{41} &= (9\mu + 2)/16, \\ \ell_{42} &= -(9\mu + 14)/32, & q_3 &= (53\mu - 26)/24, & q_4 &= -(45\mu - 17)/32. \end{aligned}$$

Define

$$\begin{aligned} \gamma &= 1 + v_1, & u_1 &= 1 + \ell_{21}v_1 + v_2, \\ u_2 &= \ell_{31}v_1 + \ell_{32}v_2 + v_3 + q_3v_1^2, & u_3 &= \ell_{41}v_1 + \ell_{42}v_2 + v_4 + q_4v_1^2, \\ x &= \gamma u_1, & y &= \gamma^2 u_2, & \theta &= \gamma^3 u_3, & C &= \rho\gamma. \end{aligned} \quad (2.4)$$

The map to be studied is the following polynomial extension:

$$\mathcal{F} = \left(C, \frac{S_1}{C}, \frac{S_2}{C^2}, \frac{S_3}{C^3}, \frac{S_4}{C^3} \right). \quad (2.5)$$

Here the substitutions (2.4) precede division.

Lemma 2.1 (Finite coefficient identities). *After the substitutions above, S_1, S_2, S_3, S_4 are divisible in $k[\rho, Y_1, Y_2, Y_3, Y_4]$ by C, C^2, C^3, C^3 , respectively. The quotient coordinates of (2.5) have term counts $(2, 207, 273, 343, 372)$ including C . Furthermore*

$$\det \frac{\partial(S_1, S_2, S_3, S_4)}{\partial(\gamma, x, y, \theta)} = \gamma^2. \quad (2.6)$$

Proof. These are exact polynomial identities over the quadratic number field, and constitute the finite-arithmetic part of the construction. Expand (2.1)–(2.4), reduce each coefficient by $9\mu^2 - 4\mu + 4 = 0$, and divide each of the four numerators by its stated power of $\rho(1 + \rho Y_1)$. All four remainders are zero. Multiplication by the four divisors recovers the numerators coefficientwise. Expansion and differentiation of the four expressions (2.2) gives (2.6). The accompanying exact-arithmetic supplement performs these finite operations and binds all quotients to the complete coefficient list. Thus this lemma uses polynomial coefficient checks, not evaluations at finitely many points. $\square$

Proposition 2.2. *The map (2.5) is polynomial and $\det J(\mathcal{F}) = 1$.*

Proof. Polynomiality follows from Lemma 2.1. On $\rho\gamma \neq 0$, the determinants of the weighted source layer, triangular layer, weighted parameter layer, radial sweep, and target division layer are, in order,

$$\rho^9, \quad 1, \quad \gamma^6, \quad \gamma^3, \quad (\rho\gamma)^{-9}.$$

The radial sweep contributes $\gamma \det J(S) = \gamma^3$, because $C = \rho\gamma$ and the other outputs are independent of ρ . Their product is one. The determinant of the polynomial extension is a polynomial; equality on this dense open proves the identity everywhere. $\square$

3. Quartic fibers and geometric degree

For independent sweep target coordinates (S_1, S_2, S_3, S_4) write

$$\begin{aligned} A &= s_2 - S_1, & B &= s_1 + 2S_2, & D &= s_0 - S_3, \\ \Delta &= B^2 - 4AD, & L &= a_2D - \frac{1}{2}a_1B + a_0A + \kappa - S_4, \\ T &= Ax^2 + Bx + D, & N &= Ax + B/2. \end{aligned} \quad (3.1)$$

Then $N^2 - AT = \Delta/4$. The first three sweep equations give

$$\tau = T, \quad \theta = 2xS_2 - x^2S_1 - S_3, \quad z = N/T, \quad y = z - a(x). \quad (3.2)$$

The fourth equation is equivalent on $T \neq 0$ to

$$P(x) = LT^2 + \mu N^3 - ATN = 0. \quad (3.3)$$

Writing $\beta = 3\mu + 2$, recovery of the parameter is

$$\gamma = \frac{\Gamma}{2T}, \quad \Gamma = -2AT + \beta N^2. \quad (3.4)$$

These identities follow by substituting (2.1); no root approximation or denominator-clearing implication in the reverse direction is needed on $T \neq 0$.

Proposition 3.1. *The sweep S and the map $\mathcal{F}$ both have geometric degree four.*

Proof. Equation (2.6) implies dominance in characteristic zero. Set $K_0 = k(S_1, S_2, S_3)$. Since T has nonzero discriminant over K_0 , $\gcd(T, N) = 1$. Write $P = -S_4T^2 + Q$, with $Q \in K_0[x]$. Modulo T one has $Q \equiv \mu N^3$, so $\gcd(T^2, Q) = 1$. Thus P is primitive and linear in S_4 in $K_0[x][S_4]$ and is irreducible there. Its degree in x is four, with leading coefficient $LA^2 \neq 0$; hence it is not associated to a polynomial in $K_0[S_4]$. It remains irreducible in $K_0(S_4)[x]$. The same argument works after algebraic extension of the constants. Formula (3.2) shows that the sweep function-field extension is generated by x , so it has geometric degree four. Generically $T = 0$ cannot occur in a fiber because it would imply $N = 0$, contrary to $\gcd(T, N) = 1$.

Every source layer in (2.3)–(2.4) is birational on the indicated nonzero-coordinate open. Explicitly, the triangular inverse is

$$\begin{aligned} v_1 &= \gamma - 1, & v_2 &= u_1 - 1 - \ell_{21}v_1, \\ v_3 &= u_2 - \ell_{31}v_1 - \ell_{32}v_2 - q_3v_1^2, & v_4 &= u_3 - \ell_{41}v_1 - \ell_{42}v_2 - q_4v_1^2. \end{aligned}$$

Recover u_i by division by $\gamma, \gamma^2, \gamma^3$, and recover Y_i by division by $\rho, \rho^2, \rho^3, \rho^3$. Finally $\rho = C/\gamma$. Consequently

$$\begin{aligned} k(\rho, Y_1, Y_2, Y_3, Y_4) &= k(C, \gamma, x, y, \theta), \\ k(\mathcal{F}) &= k(C, S_1, S_2, S_3, S_4). \end{aligned}$$

The variable C is independent of the sweep coordinates. Adjoining it does not change their finite extension degree, proving the assertion. $\square$

3.1. An exact four-point fiber

Specialize the sweep target to $(1, 0, 0, 0)$ and put

$$p(t) = 9t((219\mu + 26)t^3 + (301\mu - 170)t^2 + (45\mu - 74)t + (63\mu - 46)). \quad (3.5)$$

In this specialization $P(t) = -p(t)/2304$. In $R = k[t]/(p)$ use (3.1)–(3.4) with $x = t$, and define $\rho = 1/\gamma$. Recover u_i, v_i, Y_i by the inverse formulas in the proof of Proposition 3.1. This gives a single residue tuple $\mathcal{P} \in R^5$ encoding the four points.

Proposition 3.2. *All divisions in this construction are by units of R . The polynomial p is squarefree, and its four roots yield four distinct points $\mathcal{P}_i$ satisfying $\mathcal{F}(\mathcal{P}_i) = (1, 1, 0, 0, 0)$.*

Proof. Writing $p = tg$, exact calculation gives

$$g(0) = -288 + 252\eta \neq 0, \quad \text{disc}(g) = -6714940538880 - 1273868550144\eta \neq 0.$$

Thus p is squarefree. The Euclidean algorithm applied to the displayed quadratic T , quadratic Γ and quartic p gives

$$\gcd(p, T) = \gcd(p, \Gamma) = 1.$$

These small exact gcds, or their Bézout identities, can be checked in the supplement. Hence T , $\gamma = \Gamma/(2T)$ and $\rho = 1/\gamma$ are units. Equations (3.2)–(3.3) give the sweep target, and (2.5) with $C = 1$ gives the stated final target. To verify distinctness, the recovered source coordinates satisfy

$$t = (1 + \rho Y_1)(1 + \ell_{21}\rho Y_1 + \rho^2 Y_2).$$

Equal source tuples would therefore give equal roots. All four roots are distinct. The full canonical coefficient list is also evaluated at the same residue tuple by the exact checker, giving the same five outputs. $\square$

Both complex base changes are noninjective: an embedding of k into $\mathbb{C}$ extends to the finite field containing the four algebraic points, and preserves distinctness and polynomial evaluation. In conjunction with Proposition 2.2, this gives a noninjective Keller map in dimension five. The contribution studied below is its specific geometry, not priority for the Jacobian Conjecture. The collision alone asserts at least four points; the generic degree was proved separately.

4. A component of the nonproperness set

Set

$$K = -\frac{24\mu(\mu + 2)^2}{\beta^4} = \frac{-7 + 4\eta}{12}, \quad H_0 = \Delta L^2 - KA^4. \quad (4.1)$$

The quartic shell is controlled by the following identities:

$$\begin{aligned} \text{lc}_x(P) &= A^2 L, \\ \text{Res}_x(P, T) &= -A^4 \mu^2 \Delta^3 / 64, \\ \text{Res}_x(P, \Gamma) &= A^4 \Delta^3 \beta^4 H_0 / 256, \\ \text{Disc}_x(P) &= -\mu^2 A^4 \Delta^3 H_0^2 / 4. \end{aligned} \quad (4.2)$$

They follow by expanding the Sylvester determinants and reducing by $9\mu^2 - 4\mu + 4 = 0$. The supplement checks each coefficient of these identities. For the smaller resultant, one can also evaluate $P = \mu N^3$ at the two roots of T and use $N^2 = \Delta/4$, with product of the two N -values equal to $-\Delta/4$.

Lemma 4.1. *The polynomial $H_0 \in k[A, B, D, L]$ is absolutely irreducible.*

Proof. As a polynomial in D it is $(-4AL^2)D + (B^2L^2 - KA^4)$. The two coefficients are coprime in $k[A, B, L]$: neither A nor L divides the constant coefficient. It is therefore primitive and linear in D . Gauss's lemma applies over any algebraic extension of k . $\square$

On $H_0 = 0$ and $AL \neq 0$, write

$$N_0 = \frac{1 - \eta}{4} \frac{A^2}{L}, \quad N_1 = \frac{1 + 11\eta}{36} \frac{A^2}{L}.$$

Using $\Delta = KA^4/L^2$ gives

$$\begin{aligned} A^2P &= L(N - N_0)^3(N - N_1), \\ N_1 - N_0 &= \frac{-2 + 5\eta}{9} \frac{A^2}{L}, \\ T(N_0) &= \frac{2 - 5\eta}{24} \frac{A^3}{L^2}, & T(N_1) &= \frac{-26 - 43\eta}{648} \frac{A^3}{L^2}, \\ \Gamma(N_0) &= 0, & \Gamma(N_1) &= \frac{-112 - 17\eta}{243} \frac{A^4}{L^2}. \end{aligned} \quad (4.3)$$

Here $T(N) = (N^2 - \Delta/4)/A$; each identity follows by direct substitution. The quartic has one triple and one simple root, with $\gamma = 0$ and $\gamma \neq 0$, respectively. Neither has $T = 0$. The open is nonempty: $(A, B, D, L) = (1, 0, (7 - 4\eta)/48, 1)$ lies in it.

4.1. The final target equation

Use final target coordinates (C, Z_1, Z_2, Z_3, Z_4) and substitute

$$(S_1, S_2, S_3, S_4) = (CZ_1, C^2Z_2, C^3Z_3, C^3Z_4). \quad (4.4)$$

Let $\widehat{H}_0(S)$ be (4.1) after the affine-linear substitution (3.1). Direct collection of powers of C gives a polynomial h determined by

$$\widehat{H}_0(CZ_1, C^2Z_2, C^3Z_3, C^3Z_4) = C^3h(C, Z). \quad (4.5)$$

Its boundary value is

$$\begin{aligned} h(0, Z) &= -\frac{1}{4}Z_1^3 + \frac{1+2\eta}{4}Z_1Z_2 + \left(\frac{7}{12} - \frac{\eta}{3}\right)Z_3 \\ &\quad + \left(\frac{9}{64} + \frac{9\eta}{256}\right)Z_4. \end{aligned} \quad (4.6)$$

In particular

$$\partial_{Z_4}h|_{C=0} = 9(4 + \eta)/256 \neq 0. \quad (4.7)$$

Equations (4.5)–(4.6) are also finite exact coefficient identities, included in the supplement. They define h without relying on an unprovided expanded equation.

Lemma 4.2. *The polynomial h is absolutely irreducible. For $X = V(h)$ one has*

$$X \cap D(C) \cong \mathbb{G}_m \times V(H_0), \quad X \cap V(C) \cong \mathbb{A}^3.$$

The second locus consists entirely of smooth points of X .

Proof. Over $k[C, C^{-1}]$ the affine-linear change from Z to (A, B, D, L) has determinant $-2C^9$ and a triangular inverse. It identifies h , up to a unit, with H_0 . Lemma 4.1 implies localized absolute irreducibility. Any factor becoming a unit after localization is a scalar times a power of C ; but $C \nmid h$ by (4.6). Hence h is absolutely irreducible. The first isomorphism is the same coordinate change. Formula (4.6) solves uniquely for Z_4 on the boundary, giving the second; the nonzero derivative (4.7) proves smoothness there. $\square$

Proposition 4.3. *The fourfold $X = V(h)$ is an irreducible component of $J(\mathcal{F})$.*

Proof. For $C \neq 0$ the final fiber is in bijection with the sweep fiber over (4.4) after deleting $\gamma = 0$. Indeed a finite final preimage has $C = \rho\gamma \neq 0$, and all inverse coordinate layers are defined. Conversely any sweep point with $\gamma \neq 0$ recovers $\rho = C/\gamma$ and all source coordinates uniquely.

The nonempty open in (4.3), transported through Lemma 4.2, is dense in X . At every point of this open the triple sweep root has $\gamma = 0$ and no finite final lift; the simple root has exactly one. Thus the final fiber has just one distinct point. Such a target cannot lie outside $J(\mathcal{F})$. Otherwise the map would be finite over a connected neighborhood and, by Proposition 2.2, finite étale. Its locally free rank would be the generic degree four from Proposition 3.1, so every geometric fiber would consist of four reduced points.

The nonproperness set is closed [5]; equivalently it is the target image of the boundary in the projective closure of the graph. It contains the closure X of the dense one-point-fiber open. It is a proper closed subset of $\mathbb{A}^5$ because $\mathcal{F}$ is generically finite. Any irreducible component containing the irreducible fourfold X has dimension four, and hence equals X . $\square$

5. The intrinsic singular geometry

Proposition 5.1. *The reduced singular locus of X is the union of two components $\mathbb{G}_m \times \mathbb{A}^2$ and $\mathbb{G}_m \times \mathbb{A}^1$, with intersection $\mathbb{G}_m$. Moreover $[X] = \mathbb{L}^4$ and $E_c(\text{Sing}(X)_{\text{red}}; u, v) = (uv)^3 - 2uv + 1$.*

Proof. The four derivatives of H_0 are

$$-4DL^2 - 4KA^3, \quad 2BL^2, \quad -4AL^2, \quad 2(B^2 - 4AD)L.$$

If $L = 0$, the equation $H_0 = 0$ forces $A = 0$. If $L \neq 0$, vanishing of the D , B , and A derivatives forces $A = B = D = 0$. The converses also hold, so

$$\text{Sing } V(H_0)_{\text{red}} = \{A = L = 0\} \cup \{A = B = D = 0\}.$$

The two components are $\mathbb{A}^2$ and $\mathbb{A}^1$ meeting at the origin. The open isomorphism in Lemma 4.2 multiplies this description by $\mathbb{G}_m$; there are no singular boundary points. These are also the global singular components: their closures cannot acquire smooth points, since the singular locus is closed. Inclusion–exclusion therefore gives, with $q = uv$,

$$E_c(\text{Sing}(X)_{\text{red}}) = (q - 1)q^2 + (q - 1)q - (q - 1) = q^3 - 2q + 1.$$

For completeness, stratify $V(H_0)$ by $L = 0$, by $L \neq 0, A = 0$, and by $LA \neq 0$. These strata are respectively $\mathbb{A}^2$, $\mathbb{G}_m \times \mathbb{A}^1$, and $\mathbb{G}_m^2 \times \mathbb{A}^1$: in the last case solve uniquely for D . Thus

$$[V(H_0)] = \mathbb{L}^2 + (\mathbb{L} - 1)\mathbb{L} + (\mathbb{L} - 1)^2\mathbb{L} = \mathbb{L}^3.$$

Lemma 4.2 now gives $[X] = (\mathbb{L} - 1)\mathbb{L}^3 + \mathbb{L}^3 = \mathbb{L}^4$. The Hodge realization yields $E_c(X) = q^4$. $\square$

6. Normalized profiles and stable dimension

The following general observation states exactly what the singular calculation detects. We recall the compact-support Hodge–Deligne polynomial E_c with its additivity and multiplicativity, and $E_c(\mathbb{A}^1) = uv$. These follow from mixed Hodge structures on compact-support cohomology [2].

Lemma 6.1 (Common stabilization). *If $F : \mathbb{A}^n \rightarrow \mathbb{A}^n$ and $G : \mathbb{A}^m \rightarrow \mathbb{A}^m$ satisfy $F \approx G$, there is $N \geq n, m$ such that $F \times \text{id}_{\mathbb{A}^{N-n}}$ and $G \times \text{id}_{\mathbb{A}^{N-m}}$ are polynomially left–right equivalent.*

Proof. Choose a finite generating zigzag and let N be its maximum dimension. Pad every map and every automorphism by identities. An extension edge then becomes equality up to a coordinate permutation; an equivalence edge remains an equivalence. Compose the padded equivalences. $\square$

Source and target isomorphisms preserve finiteness neighborhoods, and identity products satisfy

$$J(\beta \circ F \circ \alpha) = \beta(J(F)), \quad J(F \times \text{id}_{\mathbb{A}^r}) = J(F) \times \mathbb{A}^r. \quad (6.1)$$

For the product identity one may equivalently use escaping sequences over $\mathbb{C}$: convergence of the added output coordinates forces the added input coordinates to stay bounded. These are background identities, consistent with the normalization-boundary formulation in [11].

Let Φ assign to a reduced complex variety an actual constructible variety, with

$$Z \cong Z' \Rightarrow \Phi(Z) \cong \Phi(Z'), \quad \Phi(Z \times \mathbb{A}^1) \cong \Phi(Z) \times \mathbb{A}^1.$$

A constructible variety here means a finite disjoint union of reduced locally closed varieties, considered up to piecewise algebraic isomorphism. The principal example is $\Phi(Z) = \text{Sing}(Z)_{\text{red}}$. For $q = uv$ define the multiset

$$\mathcal{E}_\Phi(F) = \{\{q^{-n} E_c(\Phi(Z); u, v) : Z \in \text{Irr } J(F)\}\} \subset \mathbb{Z}[u^{\pm 1}, v^{\pm 1}]. \quad (6.2)$$

Equal entries retain their multiplicities.

Theorem 6.2 (Normalized component profiles). *If $F \approx G$, then $\mathcal{E}_\Phi(F) = \mathcal{E}_\Phi(G)$. More precisely, components $Z \subset J(F)$ and $W \subset J(G)$ can be matched so that, for a common N ,*

$$\Phi(Z) \times \mathbb{A}^{N-n} \cong \Phi(W) \times \mathbb{A}^{N-m}. \quad (6.3)$$

Consequently the multiset of localized Grothendieck classes $\{\{\mathbb{L}^{-n}[\Phi(Z)]\}\}$ is also invariant in $K_0(\text{Var}_{\mathbb{C}})[\mathbb{L}^{-1}]$.

Proof. Lemma 6.1 and (6.1) match components by actual isomorphisms $Z \times \mathbb{A}^{N-n} \cong W \times \mathbb{A}^{N-m}$. Apply Φ and its product property to obtain (6.3). Taking classes and multiplying by $\mathbb{L}^{-N}$ only in the localization gives the motivic assertion. Taking Hodge polynomials and multiplying by q^{-N} gives the claimed profile equality. No cancellation theorem is used. $\square$

For a nonzero $P \in \mathbb{Z}[u, v]$ define

$$\text{ord}_{uv} P = \max\{r \geq 0 : P \in (uv)^r \mathbb{Z}[u, v]\}.$$

This is bivariate divisibility, not an assumption that P depends only on uv . The Hodge polynomial of a nonempty reduced variety V is nonzero: if $d = \dim V$, its $u^d v^d$ coefficient is the number of top-dimensional irreducible components. Indeed the weights on $H_c^i(V)$ are at most i , so a type (d, d) term occurs only in $H_c^{2d}(V)$, where the top-dimensional fundamental classes each contribute $\mathbb{Q}(-d)$ with positive sign.

Corollary 6.3 (Dimension obstruction). *If $Z \subset J(F)$ has $\Phi(Z) \neq \emptyset$ and $\nu = \text{ord}_{uv} E_c(\Phi(Z))$, then every m -dimensional map $G \approx F$ satisfies $m \geq n - \nu$.*

Proof. The matched profile identity is $q^{-n} E_c(\Phi(Z)) = q^{-m} E_c(\Phi(W))$. Taking the minimum of the smallest u - and v -exponents gives $\nu - n = \text{ord}_{uv} E_c(\Phi(W)) - m \geq -m$. $\square$

Completion of Theorem 1.1. The first three assertions are Propositions 2.2, 3.1, 3.2, 4.3, and 5.1. Suppose $X \times \mathbb{A}^a \cong W \times \mathbb{A}^b$ with $b > a$. Reduced singular loci commute with smooth affine factors, so their Hodge polynomials would satisfy

$$(q^3 - 2q + 1)q^a = E_c(\text{Sing}(W)_{\text{red}})q^b.$$

Cancel q^a in the integral domain $\mathbb{Z}[u, v]$. The right-hand side is divisible by q , whereas the left-hand side is not. This contradiction proves the cylinder assertion. Finally use $\Phi = \text{Sing}(-)_{\text{red}}$ in Corollary 6.3. Proposition 4.3 places the computed X in $J(\mathcal{F})$ and Proposition 5.1 gives $\nu = 0$; hence $m \geq 5$. $\square$

Remark 6.4 (An elementary check). For direct products, one can avoid Hodge theory. If $X \cong W \times \mathbb{A}^1$, the two irreducible singular components must be products of the two components of $\text{Sing}(W)_{\text{red}}$. Their intersection would be an affine-line cylinder. Its dimension is one and it is irreducible, so it would be $\mathbb{A}^1$. The actual intersection is $\mathbb{G}_m$, whose coordinate ring has a nonconstant unit. This contradicts $\mathbb{G}_m \cong \mathbb{A}^1$. The normalized profile extends this obstruction to a common stabilization in any larger dimension.

7. Reproducibility and scope

The supplementary coefficient list defines a polynomial map by 1,197 monomial records in the variable order $(\rho, Y_1, Y_2, Y_3, Y_4)$, with coefficients $a + b\eta \in k$. Its SHA-256 is

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ace8f69bc55d219d4c25901f866a3109
97a128726d4e0dbdce9a533687855ee0
```

The two displayed lines concatenate to one digest. It identifies the exact coefficient data, not a proof or an external certification. The version-matched supplement includes construction binding, global-determinant arithmetic, exact collision evaluation, and shell identities. Its README specifies the commands, required runtimes and coverage. These computations operate over rational or finite-field coefficient rings; no floating-point inference is used for the theorems. The geometric arguments proving irreducibility, component membership, the singular description and profile invariance are given in the text.

The profile equality is a necessary invariant, not a classification of stable equivalence. The total motivic calculation is deliberately included as a negative control: $[X] = \mathbb{L}^4$ alone yields no obstruction here. The map-specific construction and obstruction are the results of this paper; the standard product identities, mixed Hodge theory, and existence of noninjective Keller maps are not presented as original general tools.

Manuscript information

Astrée Iriart is an AI research system working under human supervision. This manuscript has been checked by AI, but has not been independently verified by a human or externally peer reviewed. The dates on the title page record when the results were obtained and when this version was prepared; they do not assert a public priority date. The public edition consists of this PDF and a verification-only supplement carrying the same paper identifier and version. The supplement

contains coefficient data, standalone checkers, instructions and checksums, not manuscript source files.

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